

# *Betelgeuse, Betelgeuse, Betelgeuse: are you alone?*

## Abstract

Data from this collection have been obtained through ESO program 114.28H9. These are SPHERE-ZIMPOL observations of Betelgeuse. The goal of the program was to search for the close companion of Betelgeuse at its maximum elongation of December 2024.

## Overview of Observations

The observations were performed with SPHERE-ZIMPOL on Betelgeuse (at the center of the field). Two observation modes were used:

- P1: slow polarization, with the V and N\_I filters
- CI: classical imaging with pupil stabilization (hence field rotation) through the CntHa and N\_Ha filters.

In both cases, no coronagraph was used.

## Release Content

### **SPHERE\_ZIMPOL\_Betelgeuse\_P1\_N\_I\_phase3.fits**

Centered on Betelgeuse

RA/Dec (J2000): 05 55 10.30536 / +07 24 25.4304

Mode: P1

Filter: N\_I

Observing date: 2024-12-03

Seeing: 0.51 – 0.56 arcsec

Ancillary product: **SPHERE\_ZIMPOL\_Betelgeuse\_P1\_N\_I\_DOLP.fits**, degree of linear polarization

### **SPHERE\_ZIMPOL\_Betelgeuse\_P1\_V\_phase3.fits**

Centered on Betelgeuse

RA/Dec (J2000): 05 55 10.30536 / +07 24 25.4304

Mode: P1

Filter: V

Observing date: 2024-12-03

Seeing: 0.51 – 0.56 arcsec

Ancillary product: **SPHERE\_ZIMPOL\_Betelgeuse\_P1\_V\_DOLP.fits**, degree of linear polarization

### **SPHERE\_ZIMPOL\_Betelgeuse\_ADI\_CntHa\_phase3.fits**

Centered on Betelgeuse

RA/Dec (J2000): 05 55 10.30536 / +07 24 25.4304

Mode: CI

Filter: CntHa

Observing date: 2024-12-06

Seeing: 0.46 – 0.70 arcsec

Ancillary products:

**Betelgeuse\_B\_PCA\_2024-12-06\_CntHa.fits**, Principal Component Analysis residual map

**Betelgeuse\_B\_PACO\_2024-12-06\_CntHa.fits**, Angular plus Spectral Differential Imaging Patch  
Covariances signal-to-noise ratio map

### **SPHERE\_ZIMPOL\_Betelgeuse\_ADI\_N\_Ha\_phase3.fits**

Centered on Betelgeuse  
RA/Dec (J2000): 05 55 10.30536 / +07 24 25.4304  
Mode: CI  
Filter: N\_Ha  
Observing date: 2024-12-06  
Seeing: 0.46 – 0.70 arcsec  
Ancillary products:  
**Betelgeuse\_B\_PCA\_2024-12-06\_N\_Ha.fits**, Principal Component Analysis residual map  
**Betelgeuse\_B\_PACO\_2024-12-06\_N\_Ha.fits**, Angular plus Spectral Differential Imaging Patch  
Covariances signal-to-noise ratio map

## Release Notes

Photometric calibration has been performed using the American Association of Variable Stars Observers magnitude of Betelgeuse in the V and R band at the time of the observations, propagated to the various filters using a PHOENIX model for red supergiant stars (Lançon et al. 2007) at a temperature of 3700K.

To infer the photometric zeropoint (PHOTZP keyword in the FITS headers) it was used a red supergiant model spectrum for Betelgeuse, normalized to its contemporaneous V-band magnitude, and convolved with the actual ZIMPOL filter transmission curves to predict the star’s magnitude in each filter. Synthetic magnitudes are then combined with the measured aperture fluxes to derive the photometric zeropoints.

| Filter | Vega mag | AB mag   | PHOTZP AB | ABMAGLIM | ABMAGSAT |
|--------|----------|----------|-----------|----------|----------|
| CntHa  | -0.22395 | -0.03004 | 20.050435 | 10.118   | -2.030   |
| N_Ha   | -0.56919 | 0.04187  | 20.012471 | 9.115    | -1.958   |
| N_I    | -1.53734 | -1.05777 | 19.319035 | 8.338    | -3.058   |
| V      | 0.55778  | 0.57315  | 20.161580 | 10.939   | -1.427   |

## Data Reduction and Calibration

Intensity maps within this release are the direct product of the SPHERE-ZIMPOL pipeline and were produced by the High Contrast Data Centre.

The ancillary products for the P1 mode are the direct output from the processing of the Stokes images produced by the pipeline (see eg. Cannon et al. 2021 and Montargès et al. 2023).

The ancillary product for the CI mode has been obtained through high contrast processing. The Principal Component Analysis (PCA) is described in Soummer et al. 2012 and Amara & Quanz 2012. The Angular plus Spectral Differential Imaging Patch Covariances is described in Flasseur et al. 2020a,b.

## Data Format

### Files Types

This release only contains FITS images following the following naming convention:  
SPHERE\_ZIMPOL\_<source>\_<mode>\_<filter>\_phase3.fits

## Acknowledgements

Data from this release have been published in Montargès et al. (2026), A&A, 711, L12.

According to the Data Access Policy for ESO data held in the ESO Science Archive Facility, all users are required to acknowledge the source of the data with an appropriate citation in their publications.

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Based on data obtained from the ESO Science Archive Facility with DOI:  
<https://doi.org/10.18727/archive/114>

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